Ch 13: Python `tkinter` Frames, Buttons and Text I/O

Application Program Development

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Organizing Widgets with Frames

- **Frame widget**: container that holds other widgets
  - Useful for organizing and arranging groups of widgets in a window
  - The contained widgets are added to the frame widget which contains them
  - Example:
    `tkinter.Label(self.top_frame, text='hi')`

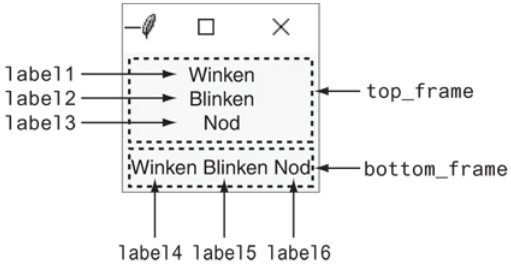


Figure 1: Arrangement of widgets using a frame (Gaddis, 2021, pg.669)

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Button Widgets

Buttons

You use the `Button` widget to create a standard button in a window. When the user clicks a button, a specified function or method is called

Callback function

A callback function is also known as an **event handler** because it handles the event that occurs when the user clicks the button.

- **`Button` widget**: widget that the user can click to cause an action to take place
  - When creating a button you can specify:
    - Text to appear on the face of the button
    - A **callback function**



Figure 2: Example of a tkinter Button (Gaddis, 2021, pg.671)

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Info Dialog Boxes

- **Info dialog box**: a dialog box that shows information to the user
  - Format for creating an info dialog box:
  - `import tkinter.messagebox` module
  - `tkinter.messagebox.showinfo(title, message)`
  - `title` is displayed in the dialog box's title bar
  - `message` is an informational string displayed in the main part of the dialog box

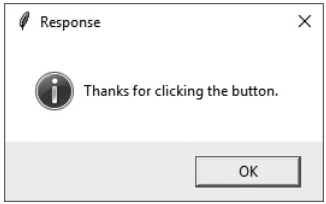


Figure 3: Example of a tkinter messagebox info dialog box (Gaddis, 2021, pg.671)

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Creating a Quit Button

- **`Quit` button**: closes the program when the user clicks it
- To create a quit button in Python `tkinter`:
  - Create a `Button` widget
  - Set the root widget's `destroy()` method as the callback function
  - When the user clicks the button the `destroy()` method is called and the program ends

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Getting Input with the Entry Widget

Entry Widget

An Entry widget is a rectangular area that the user can type input into. You use the Entry widget's get() method to retrieve the data that has been typed into the widget.

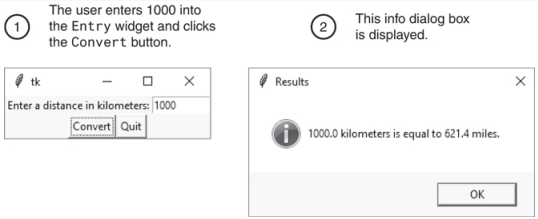


Figure 4: Example of entry widget and result dialog (Gaddis, 2021, pg.675)

- **Entry widget**: rectangular area that the user can type text into
  - Used to gather input in a GUI program
  - Typically followed by a button for submitting the data
    - The button's callback function retrieves the data from the Entry widget and processes it
  - **Entry widget's get() method**: used to retrieve the data from an Entry widget
    - Returns a string

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Using Labels as Output Fields

StringVar and Label

When a `StringVar` object is associated with a `Label` widget, the `Label` widget displays any data that is stored in the `StringVar` object.

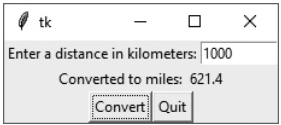


Figure 5: Example of StringVar used to dynamically display result in a Label (Gaddis, 2021, pg.678)

- Can use `Label` widgets to dynamically display output
  - Used to replace info dialog box
  - Create empty `Label` widget in main window, and write code that displays desired data in the label when a button is clocked
- **`StringVar` class**: `tkinter` module class that can be used along with a `Label` widget to display data
  - Create `StringVar` object then create `Label` widget and associate it with the `StringVar` object
  - Subsequently, any value stored in the `StringVar` object will automatically be displayed in the `Label` widget

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Summary

- A `Frame` is a container that holds other widgets. You use `Frames` in most GUI frameworks to organize the widgets in a window (GUI layout)
- A `Button` widget is a user clickable item that can be used to generate events
- A **callback function** or an **event handler** are called asynchronously by the GUI event loop when events happen.
  - For example you can associate a callback function to be called when a button press even occurs.
- Dialog boxes, like the `tkinter messagebox` are useful for popping up information for users.
- The `Entry` widget provides a basic ability to enter user text in a `tkinter` GUI application
  - Can use the `get()` method of the `Entry` object to get the text
  - Can then convert to types like `int` or `float` using built-in Python methods
- In `tkinter` a `StringVar` can be used in combination with a `Label` to dynamically display output generated by the GUI application.



References

Gaddis, T. (2021). *Starting out with Python* (5th). Pearson.

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